METHODS LEARNING NOTE

Understanding count concordance in the design-stage attrition model

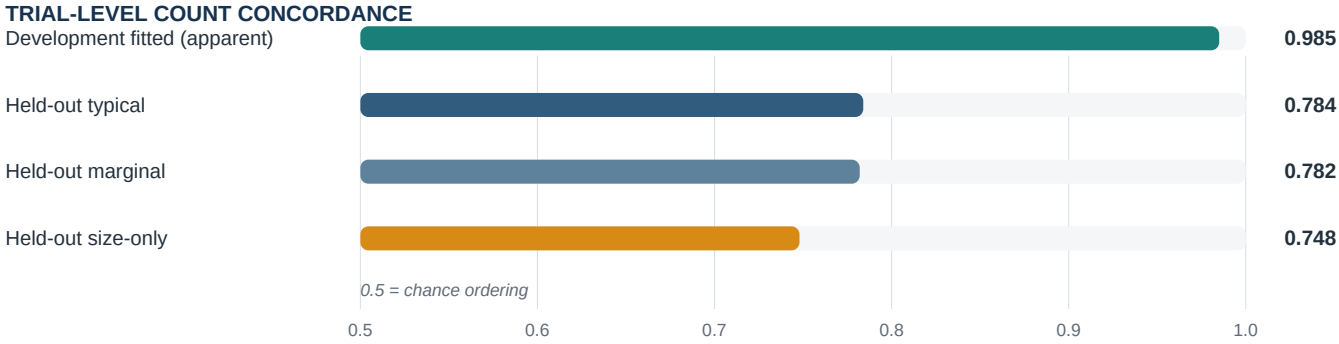
How to interpret the new trial-level measure, why it is better aligned with aggregate data, and what the apparent-versus-held-out results say about internal performance.

Bottom line

The model has excellent **apparent** capacity to rank development trials by total attrition count (concordance 0.985). In held-out trials, concordance is about 0.783: still encouraging, but the 0.202 reduction is substantial rather than slight. Trial size alone reaches about 0.748, so the full model adds a modest 0.034 to 0.036 in transportable ranking performance beyond size.

Results at a glance

Prediction	Concordance	95% bootstrap interval	What it represents
Development fitted	**0.985**	0.982 to 0.988	Apparent fit; trial-specific effects are informed by these outcomes
Held-out marginal	**0.782**	0.758 to 0.807	New-trial prediction integrating over unknown random effects
Held-out typical	**0.784**	0.759 to 0.809	New-trial prediction with random effects fixed centrally
Held-out size-only	**0.748**	0.720 to 0.773	Ranking from trial size and training-set mean risk



How to read these results

- **Apparent development performance is near-perfect.** Approximately 98.5% of comparable development-trial pairs are ordered correctly. This is a measure of fit to trials that helped estimate their own trial-specific effects, not a prospective estimate for a new trial.
- **Held-out performance is the more relevant internal-validation result.** Approximately 78% of comparable held-out pairs are correctly ordered. There is no universal adjective-based cutoff for this trial-level statistic, so the numerical estimate and comparator are more informative than calling it simply 'moderate' or 'good'.
- **Much of the ranking signal comes from trial size.** The full model improves on the size-only reference by about 3.4 to 3.6 percentage points. That increment is useful, but modest.
- **Random-effect handling barely changes rank ordering.** Marginal and typical predictions differ by only 0.002 in concordance, suggesting that integration over unknown random effects is not the main determinant of held-out ranking.

Intervals are percentile intervals from 2,000 trial-level bootstrap resamples. All estimates use 342 trials and 90 distinct observed count values.

FROM RISK RANKING TO COUNT RANKING

The old and new measures answer different questions

The old grouped C-statistic was mathematically valid for the grouped binary construction, but its participant-like estimand did not match the observed unit of analysis or the model's count-recovery target. The new measure evaluates trials directly.

Feature	Old grouped C-statistic	New count concordance
Unit being ranked	Participants represented indirectly through trial-level groups	Trials
Prediction score	Trial-level attrition risk, p[i]	Expected attrition count, N[i] x E(p[i])
Pair compared	An event participant and a non-event participant	Two trials with different observed attrition counts
Weighting	Larger trials and event/non-event totals generate more participant pairs	Each comparable unordered trial pair contributes equally
Observed ties	Within-trial event/non-event comparisons are tied because all share one score	Equal observed counts are excluded as uninformative
Predicted ties	Contribute 0.5	Contribute 0.5
Question answered	Does a randomly selected event tend to come from a higher-risk trial than a randomly selected non-event?	Does the trial with more observed attrition receive the larger expected count?
Interpretation	Grouped binary AUROC with an ecological, participant-like interpretation	Trial-level count-ranking concordance; not an individual-level AUROC

The key conceptual shift

The old question was about **who** experienced attrition, even though individual rankings could not be recovered from aggregate data. The new question is about **which trial** experienced the larger total attrition count - exactly the quantity represented in the validation data.

A simple worked example

Trial	N	Predicted risk	Expected count	Observed count
A	1,000	0.10	**100**	**95**
B	100	0.20	**20**	**22**

Trial B has the higher predicted **risk** (0.20 versus 0.10), so a risk-ranking statistic prefers B. Trial A is ten times larger and therefore has the higher predicted **count** (100 versus 20). Because A also has the higher observed count, count concordance treats this pair as concordant. Trial size therefore belongs explicitly in the count prediction.

How the new statistic is calculated

- Form every unordered pair of trials with different observed attrition counts.
- For each trial, use the posterior mean expected count: predicted count = N x E(p).
- Award 1 if the trial with the larger observed count also has the larger predicted count.
- Award 0 if the ordering is reversed, and 0.5 if the two predicted counts are tied.
- Average these scores across all comparable trial pairs. Chance ordering is 0.5; perfect ordering is 1.

What was lost by removing the grouped statistic?

Nothing essential for the stated target. The new statistic relinquishes an individual-level discrimination claim that aggregate outcomes cannot support, so the estimand now matches the data.

INTERPRETATION AND REPORTING

What the apparent-versus-held-out gap teaches us

The gap is primarily a reminder that apparent fit and new-trial prediction are different tasks. It is not evidence that the internal validation has failed.

Why the development value is so high

For the fitted development trials, the trial-specific random effects are estimated using the same attrition outcomes later used to assess concordance. They can therefore absorb trial-level departures from the population relationship. The resulting 0.985 is best labelled **apparent fitted concordance**. It describes reconstruction of the observed development trials and is expected to be optimistic for a new trial.

Why random-effect uncertainty is probably not the main issue

In cross-validation, marginal prediction integrates over unknown trial effects, whereas typical prediction fixes them at their central value. Their concordances are nearly identical (0.782 and 0.784). This suggests that the choice between these two treatments affects absolute predictions more than it affects the ordering of trials. The large apparent-versus-held-out gap is better understood as the loss of outcome-informed local effects when predicting genuinely held-out trials.

What count concordance does - and does not - show

It does show

- How reliably the model orders trials from lower to higher total attrition.
- Performance at the trial level, aligned with the aggregate validation data.
- Incremental ranking beyond a transparent size-only reference.

It does not show

- Whether predicted counts are numerically close to observed counts.
- Whether prediction intervals have appropriate coverage or sharpness.
- Individual-level risk discrimination, causality, or external validity.

Useful reporting and analysis points

- Report the held-out estimate as the main measure of internal predictive discrimination; label 0.985 as apparent.
- Report count concordance alongside count MAE, RMSE, calibration, predictive interval coverage and predictive density.
- Keep the size-only reference because expected count necessarily contains trial size.
- For formal evidence of added discrimination, bootstrap the paired difference between full-model and size-only concordance using the same trial resamples.
- Describe the measure as trial-level count concordance, not an individual-level C-statistic or AUROC.

Suggested manuscript wording

Apparent trial-level count concordance was 0.985 (95% bootstrap interval 0.982 to 0.988), although trial-specific random effects were informed by the corresponding development outcomes. In internal cross-validation, concordance was 0.782 using marginal new-trial predictions and 0.784 using typical-trial predictions, indicating correct ordering of about 78% of comparable held-out trial pairs. The trial-size-only reference achieved 0.748, showing modest additional ranking information beyond trial size. Count concordance was interpreted alongside count accuracy, calibration and predictive uncertainty.

Sources: outputs/design/posterior_count_concordance.csv and outputs/design/internal_validation/out/design_stage_cv_count_concordance.csv. Intervals use 2,000 trial resamples of fixed posterior-mean scores; they are not Bayesian credible intervals or full-refit uncertainty. Prepared 21 July 2026.